

Practice Session 2

Quiz 1

1. Using Slater's rules, calculate Z^* for a 3d electron in Mn. **3**

2. The increase in Z^* between C and N is 0.69 whereas the increase between N and O is only 0.62. Suggest a reason why the increase in Z^* for a 2p electron is smaller between N and O than between C and N.

and N.

$$(1s^2)(2s^2 2p^6)(3s^2 3p^6)(\overset{\downarrow}{3d^5})(4s^2) \quad \sigma = (4 \times 0.35) + (18 \times 1) =$$
$$Z^* = 25 - \sigma$$

Application of Slater's rule

$$\text{Na}^+ = 116$$

$$\text{O}^{2-} = 126$$

$$\text{Mg}^{2+} = 86$$

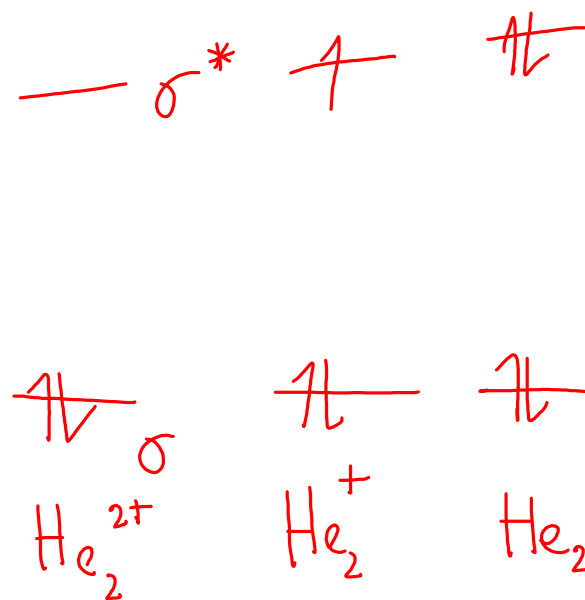
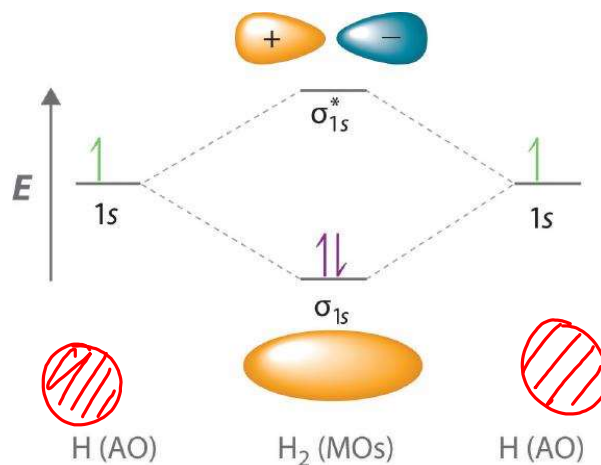
$$\text{F}^- = 119 \text{ pm}$$

1. Using Slater's rules determine Z^* for
 - a. a 3p electron in P, S, Cl. Is calculated value of Z^* consistent with the relative sizes of these atoms?
 - b. a 2p electron in O^{2-} , F^- , Na^+ and Mg^{2+} . Is the calculated value of Z^* consistent with the relative sizes of these ions?

$$\begin{array}{l|l|l}
 Z_{\text{eff}}(\text{P}) = 4.8 & \sigma = 4.15 & Z_{\text{Na}^+}^* = \\
 Z_{\text{eff}}(\text{S}) = 5.45 & Z_{\text{O}^{2-}}^* = & Z_{\text{Mg}^{2+}}^* = \\
 Z_{\text{eff}}(\text{Cl}) = 6.1 & Z_{\text{F}^-}^* = &
 \end{array}$$

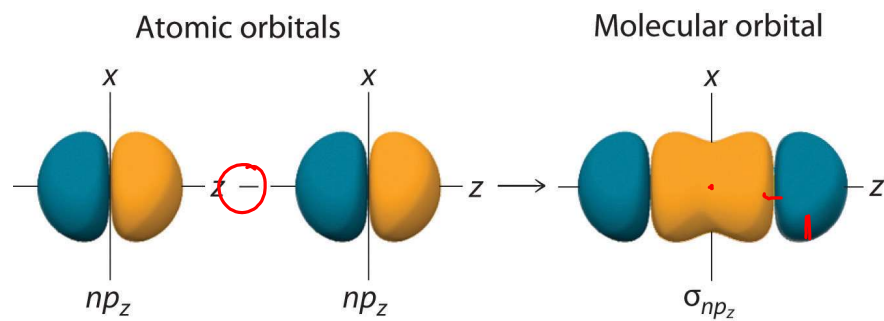
$106 \quad 102 \quad 99 \text{ (pm)}$
 $\text{P} \rightarrow \text{S} \rightarrow \text{Cl}$
 atomic radii should decrease
 $\sigma = (7 \times 0.35) + (2 \times 0.85)$
 $(1s^2)(2s^2 2p^6)$

σ - bond

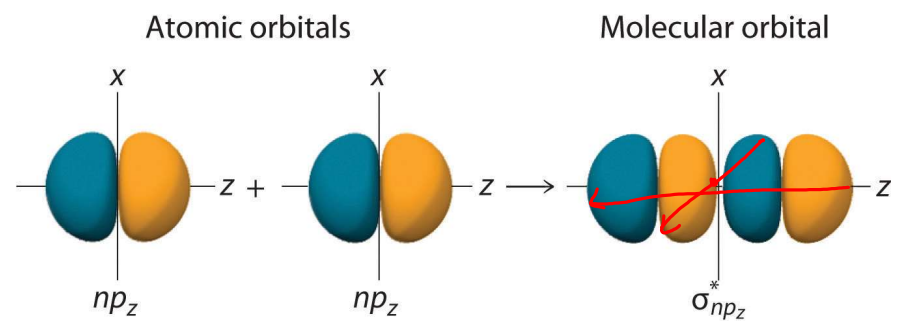
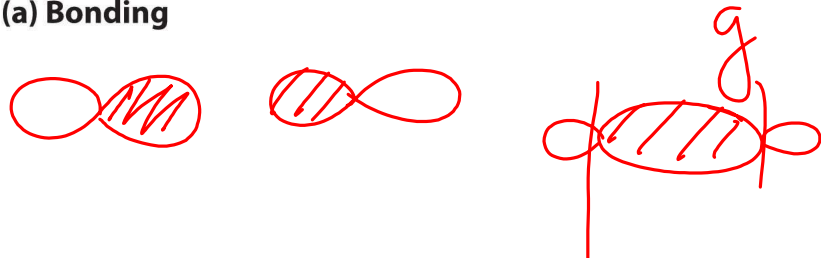


Using MOT, can you compare the bond order of He_2 , He_2^+ , and He_2^{2+} and comment on their stability?

σ - bond



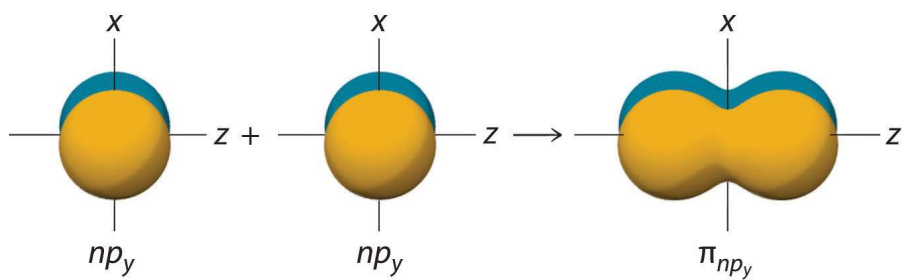
(a) Bonding



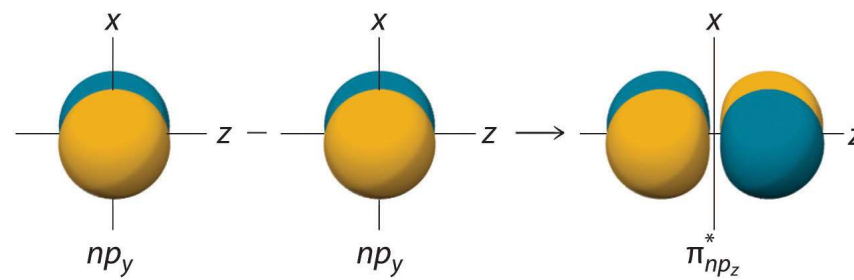
(b) Antibonding

u

π -bond

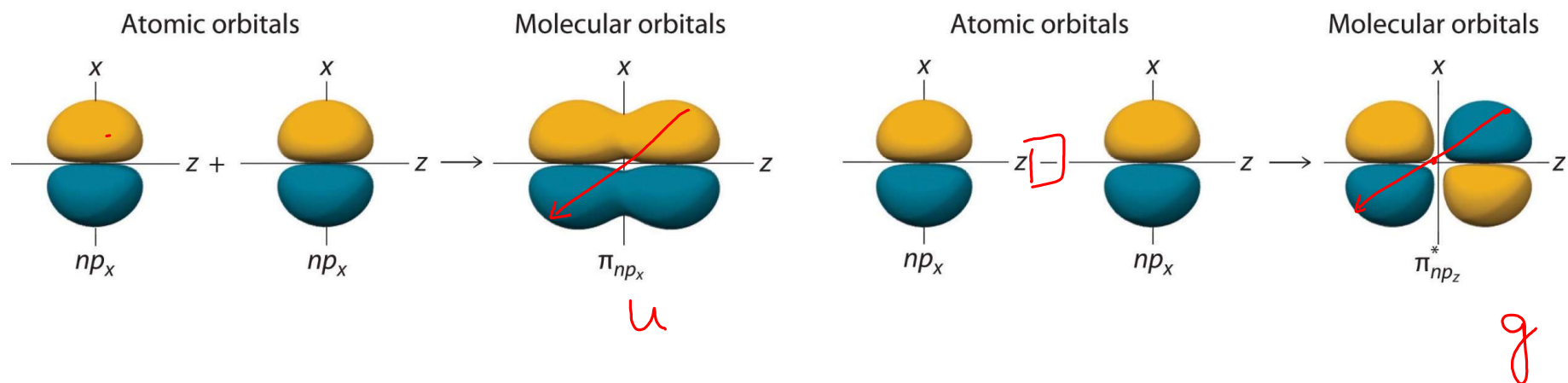


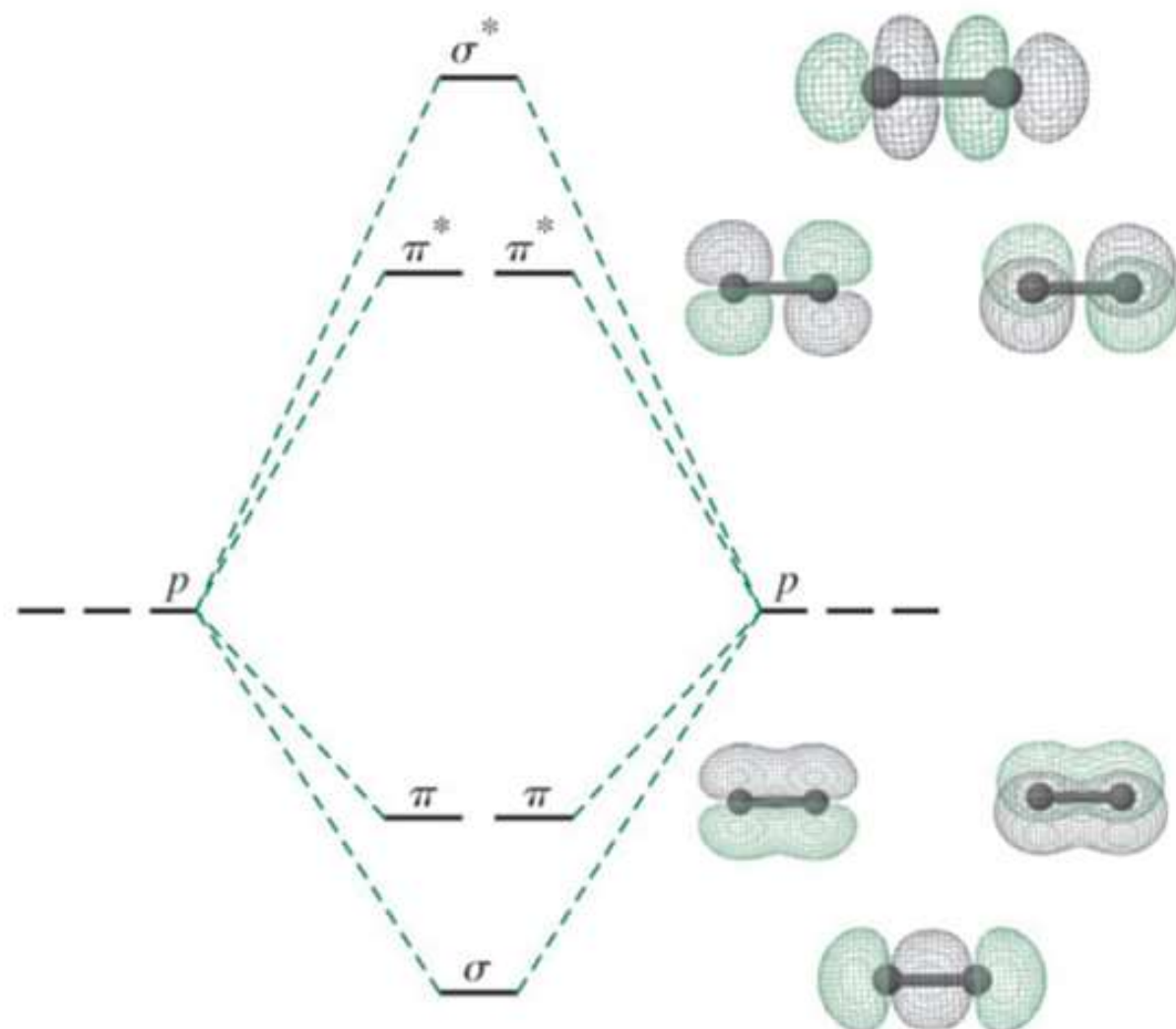
(a) Bonding



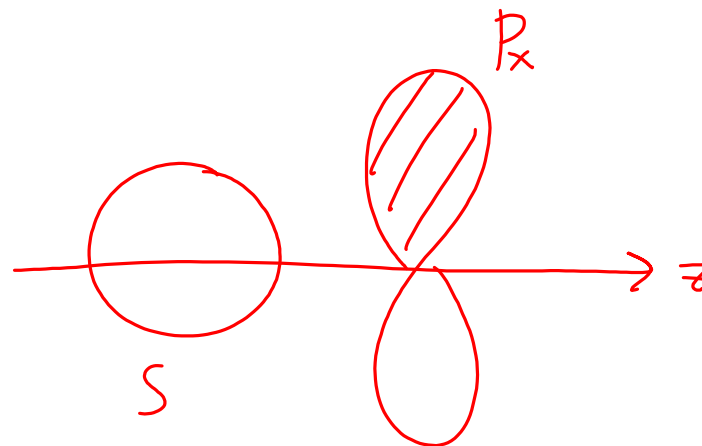
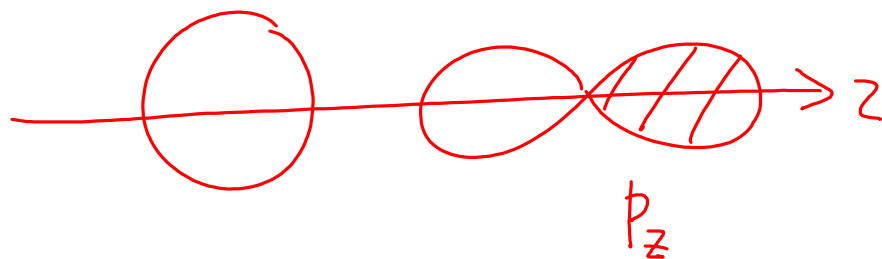
(b) Antibonding

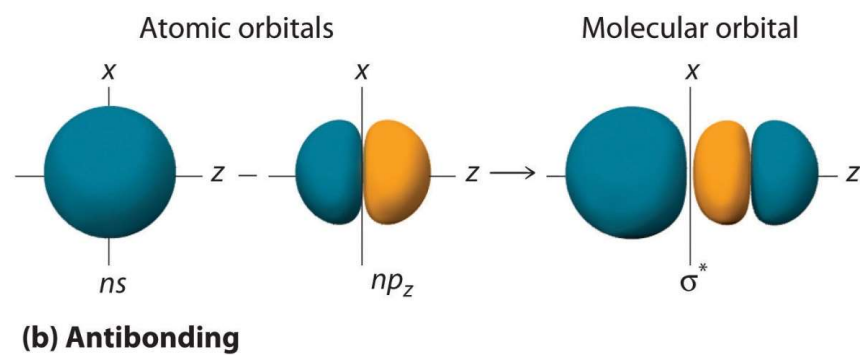
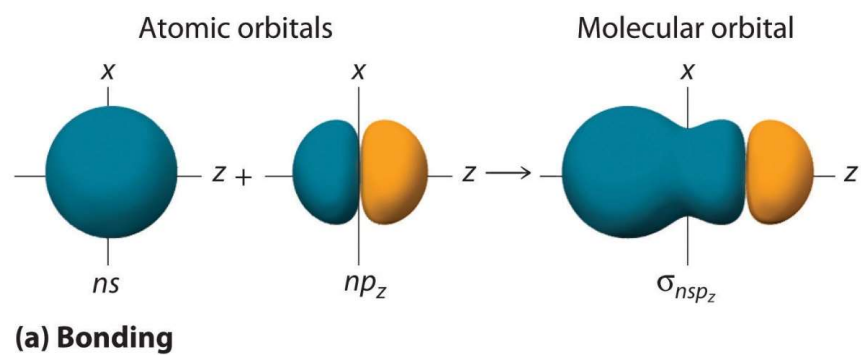
π -bond



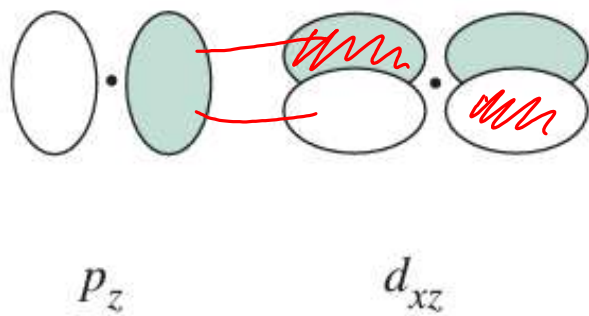


- <https://www.chemtube3d.com/salc-s-orbitaloverlap/>

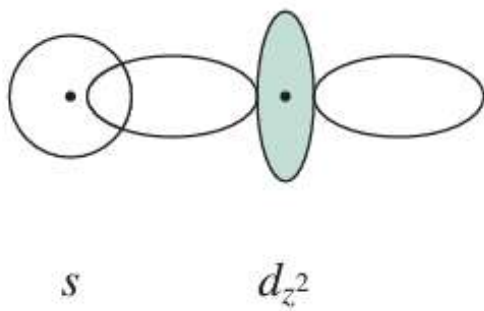




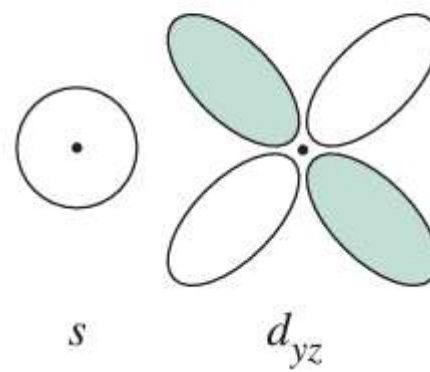
p_z and d_{xz}

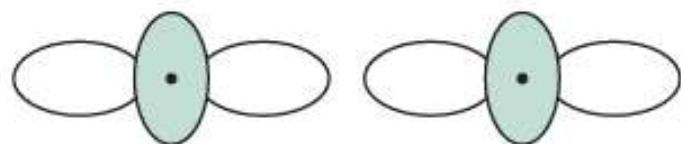


s and d_{z^2}

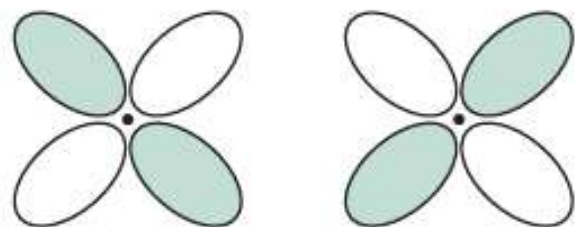


s and d_{yz}

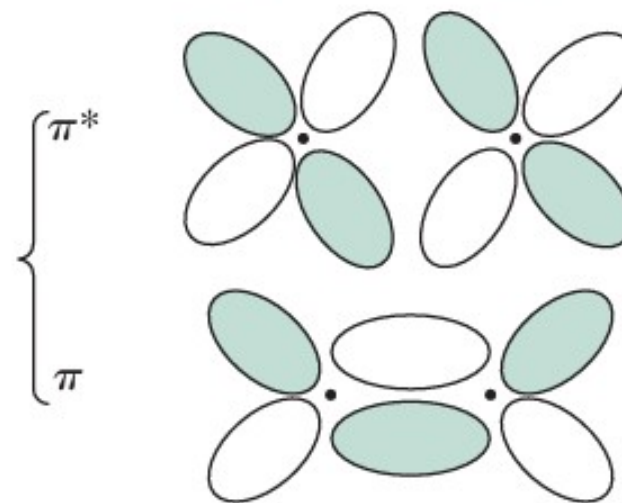
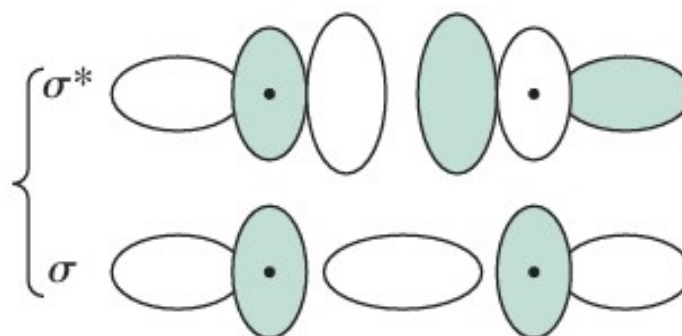


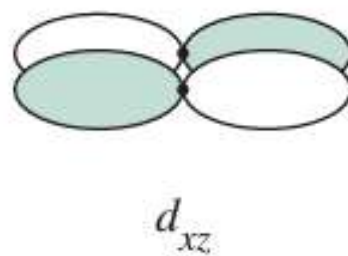
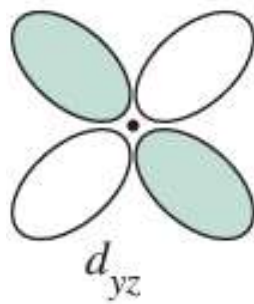
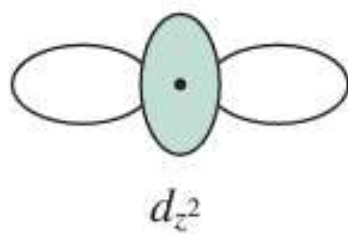
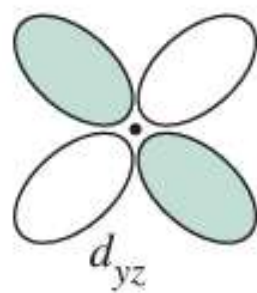


d_{z^2} orbitals
end-to-end



d_{xz} or d_{yz} orbitals
in the same plane





MOT

1. Use molecular orbital diagrams to determine the number of unpaired electrons in

a. O_2^-

b. O_2^+

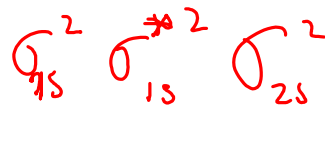
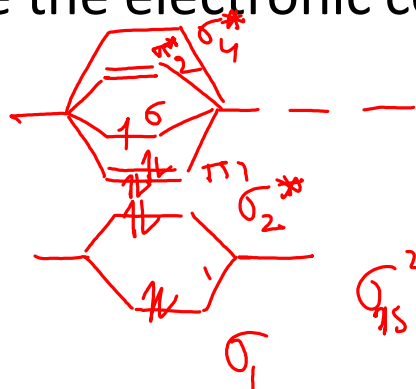
2. Write the electronic configuration of

a. Be_2

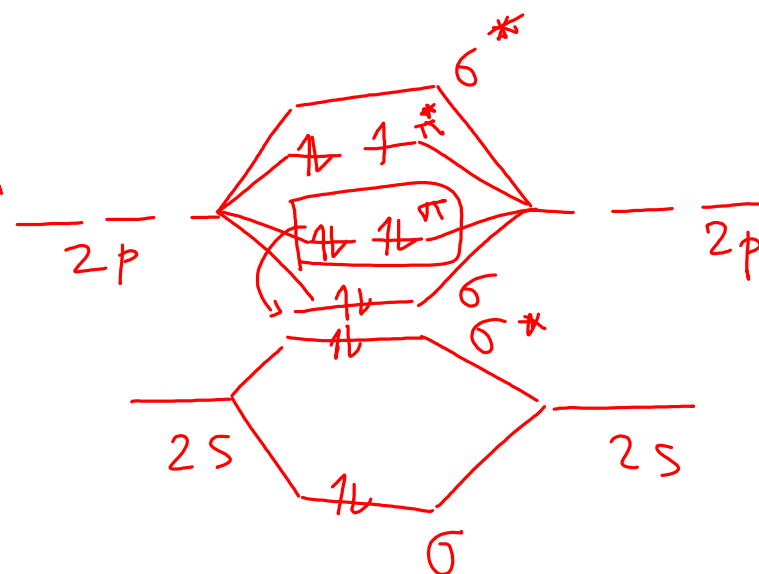
b. B_2

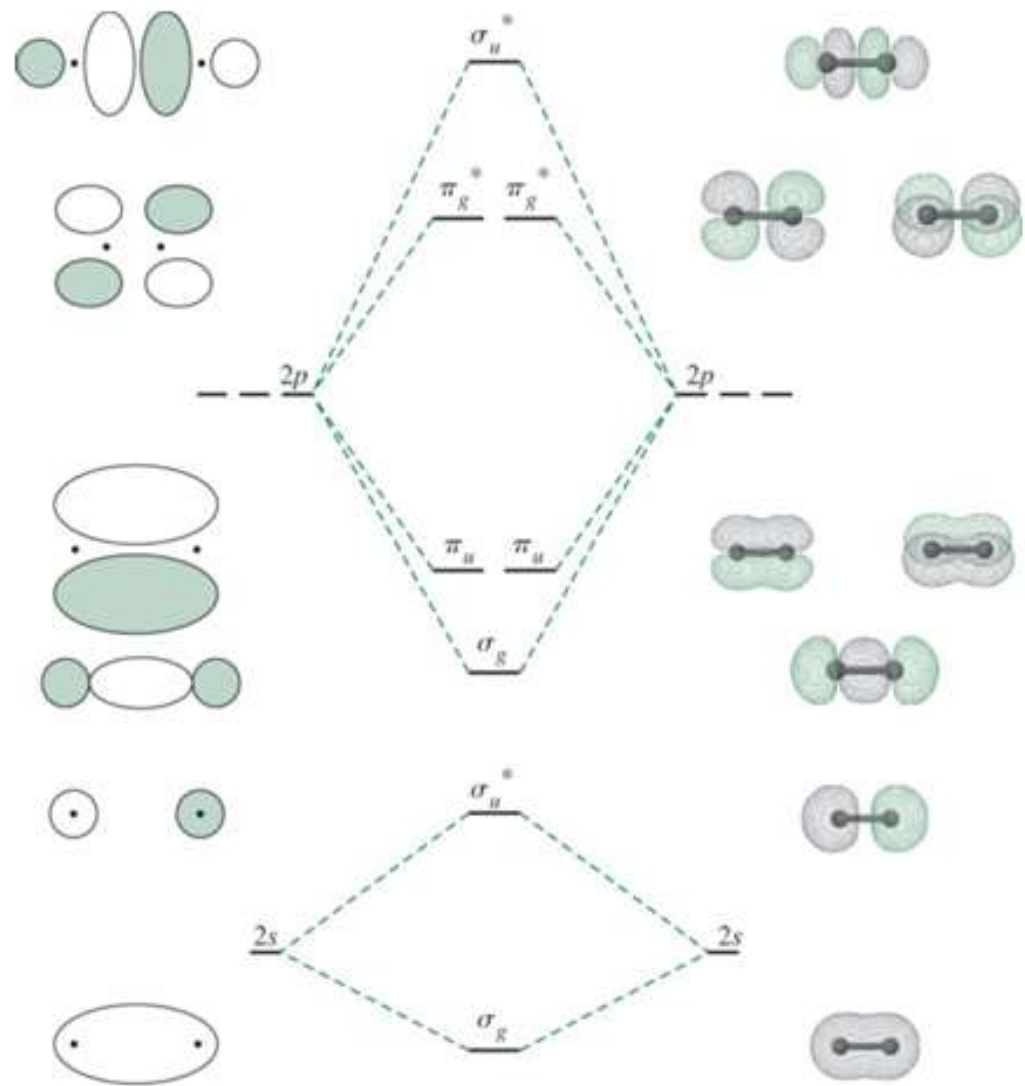
✓ c. C_2^-

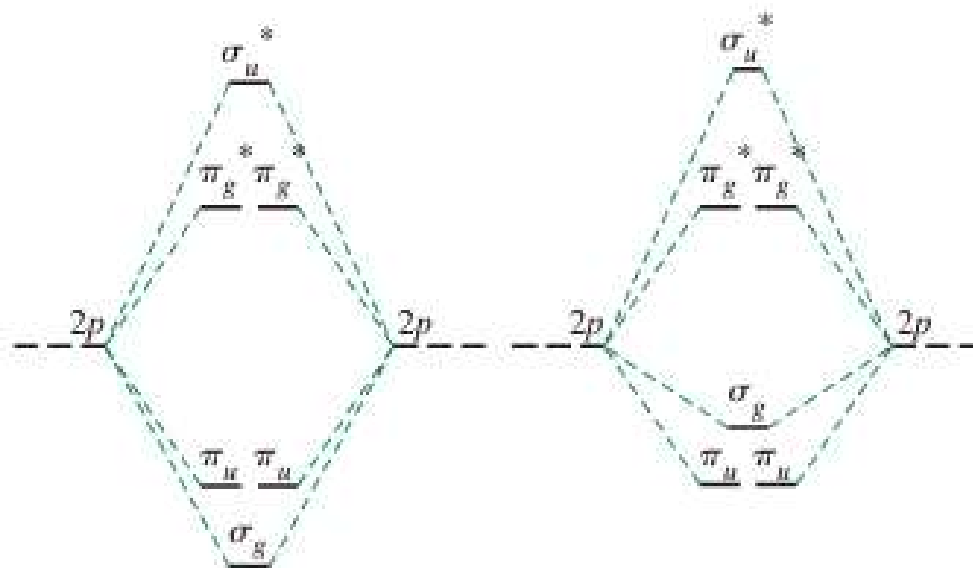
d. F_2^+



E ↑

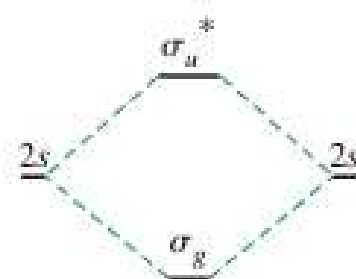






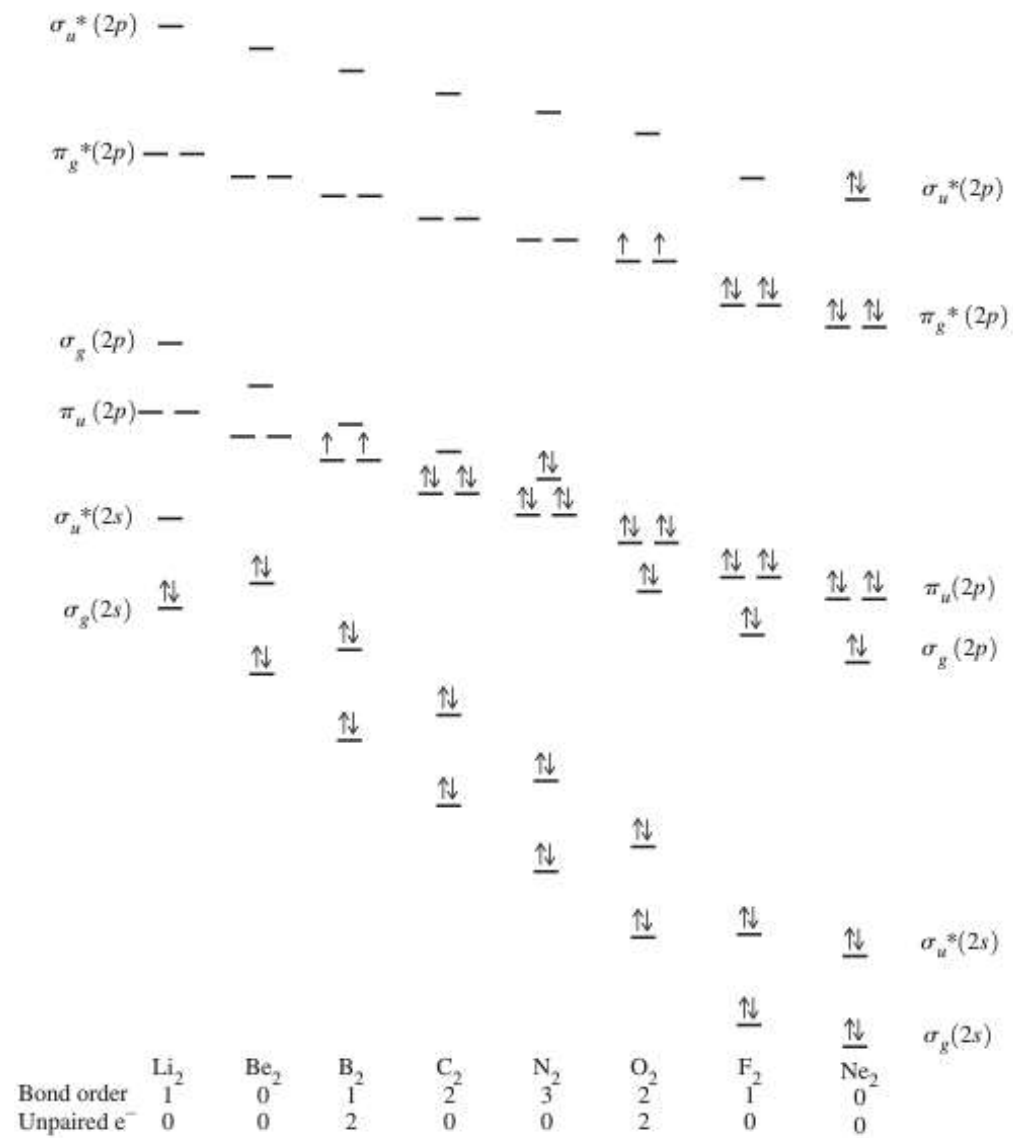
No mixing

(a)



Mixing of σ_g orbitals

(b)



3. Describe the bonding in C_2^{2-} in terms of MOT and compare the bond order to that of C_2 .

4. Using MOT, determine the bond orders of (a) S_2 (b) Cl_2 and compare the values with the bond orders determined from Lewis structure.

5. What are the expected changes in bond order and bond distance that accompany the following ionization processes?

